

Javascript

Operators, conditional statement

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Operators</title>
  </head>
  <body>
    <h1>Operators</h1>

    <script>
      // opeators => unary, binary, terinary
      //   unary => one operand

      // increment ++
      // decrement --

      let add = 10;

      // console.log(add);
      // console.log(add++);
      // console.log(++add);

      // console.log(add);

      // console.log(add--);
      // console.log(add);
      // console.log(--add);

      // binary operators => arithmetic,
      // relational, logical,
      // assignment, bitwise

      let a = 3;
      let b = 4;
      //   arithmetic

      console.log(a + b);
      console.log(a - b);
      console.log(a * b);
      console.log(a / b);
      console.log(a % b);

      //relational => > < >= <= == === != !==
```

```
console.log(a > b);
console.log(a < b);
console.log(a >= b);
console.log(a <= b);
console.log(a == b);
console.log(a != b);

console.log(a === b); // strict check
console.log(a !== b);

//logical operator => && || !

// and => true when all condition is true

console.log(true && false);
console.log(true && true);

// OR => need atleast one condition to be true

console.log(true || true);

console.log(!true);

// bitwise => & | ^ >> << ~

console.log(a & b);

console.log(a | b);

console.log(a ^ b);

console.log(a >> 2);

console.log(a << 1);

console.log(~a);

// assignment operator

let data = 10;

data += 10;
console.log(data);

// ternary

console.log(false ? console.log("true") : console.log("false"));
```

```

// conditional statement

// if , if else, if else if , nestedif , switch

if (true && true) {
    console.log("dflkaj");
}

let age = 20;

if (age >= 18) {
    console.log("You are eligible for vote ");
} else {
    console.log("you are not eligible for vote ");
}

let voterId = false;
if (age >= 18) {
    if(voterId){

        console.log("You are eligible for vote ");
    }
}

// if else if
let mark = 90;
if(mark >= 90){
    console.log("Grade A+");

}else if(mark >= 80){
    console.log("Grade B");

}else if (mark >= 70){
    console.log("Grade C");

}

// swtich statement

switch(mark){
    case 90: console.log("Grade A");
    break;
    case 80 : console.log("Grade B");
}

// a + b => operand , operators
</script>

```

```
</body>
</html>
```

Theory Notes

1. Operators

- **Unary operators** → work on a single operand.
 - ++ (increment), -- (decrement).
- **Binary operators** → work on two operands.
 - **Arithmetic:** +, -, *, /, %.
 - **Relational:** >, <, >=, <=, ==, !=, ===, !==.
 - **Logical:** && (AND), || (OR), ! (NOT).
 - **Bitwise:** &, |, ^, >>, <<, ~.
 - **Assignment:** =, +=, -=, *=, /=, %=.
- **Ternary operator** → shorthand for if-else.

```
condition ? valueIfTrue : valueIfFalse;
```

2. Conditional Statements

- **if** → executes block if condition is true.
- **if-else** → executes one block if true, another if false.
- **else-if ladder** → multiple conditions checked sequentially.
- **nested if** → `if` inside another `if`.
- **switch** → cleaner alternative for multiple discrete values.

3. Key Differences

- `==` → compares values with type coercion.
- `===` → strict comparison (checks both value and type).
- `&&` → returns true only if all conditions are true.
- `||` → returns true if at least one condition is true.
- `!` → negates a boolean value.

Coding Exercises

1. Write a program to check if a number is even or odd using the ternary operator.
2. Use relational operators to compare two strings and print whether they are equal.
3. Create a grading system using if-else-if (A+, B, C, F).
4. Demonstrate the difference between `==` and `===` with examples.
5. Write a program using `switch` to print the day of the week based on a number (1 = Monday, etc.).
6. Implement a nested `if` that checks if a person is eligible to vote (age $\geq$ 18 and has voter ID).